

Shubham Kumar Jha

Data Analyst | Aspiring Data Scientist

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SUMMARY

Data Analyst with 4+ years of quantitative research experience, specialising in business and product analytics. Skilled in Python, SQL, ETL, statistical testing, and dashboard development, with experience turning large, messy datasets into clean, decision-ready insights. Delivered end-to-end analytics work, including data cleaning, KPI reporting, and predictive modelling, across datasets ranging from 8,000 to over 500,000 rows. Known for building reliable, automated workflows and communicating findings clearly to technical and non-technical stakeholders.

SKILLS

Data Analysis & Programming Python (Pandas, NumPy, SciPy), SQL, Exploratory Data Analysis (EDA), Data Cleaning and Wrangling, Time Series Analysis, Anomaly Detection

Databases & Data Modelling MySQL, PostgreSQL, SQLite, SQL Joins, CTEs, Subqueries, Window Functions, Data Modelling, Star Schema

Visualisation & BI Power BI (DAX, Power Query), Tableau, Excel (Pivot Tables, VLOOKUP/XLOOKUP), Matplotlib, Seaborn, Plotly, Dashboard Development, Data Storytelling

Statistics & Testing Hypothesis Testing, A/B Testing, Statistical Inference, KPI Analysis, Data Validation

Machine Learning Scikit-learn, XGBoost, Classification, Regression, Feature Engineering, Cross-Validation, SHAP, ROC-AUC/PR-AUC

Tools & Cloud Git, GitHub, Streamlit, Jupyter, Linux/Bash, VS Code, AWS Fundamentals, Google Analytics

EXPERIENCE

Junior Research Fellow — Data Analytics

Aug '25 — Present

National Institute of Technology Delhi

Delhi, India

- Analyse large, multi-instrument observational datasets in Python to identify trends, anomalies, and decision-relevant patterns.
- Design and automate end-to-end ETL workflows covering extraction, data cleaning, transformation, validation, visualisation, and reporting, reducing repetitive manual work and improving reproducibility.
- Apply exploratory data analysis, correlation analysis, and time-series methods to evaluate complex, high-dimensional datasets.
- Build publication-quality visualisations and dashboards in Python/Matplotlib, and present findings clearly to multidisciplinary stakeholders.
- Validate analytical outputs against independent checks to strengthen data quality and reliability before reporting results.

Junior Research Fellow

Aug '22 — Apr '25

Indian Institute of Astrophysics

Bengaluru, India

- Processed and analysed high-volume observational data from multiple space missions using custom Python workflows.
- Built reusable data pipelines for extraction, cleaning, transformation, and visualisation, improving the reliability and repeatability of the analysis process.
- Applied statistical, correlation, and trend analysis to identify meaningful patterns in noisy, real-world data.
- Automated repetitive processing tasks and validated computational outputs against independent benchmarks to improve accuracy.
- Translated technical findings into clear visuals and insights for mixed technical and non-technical audiences.

PROJECTS

Superstore Sales, Profitability and Margin Analysis [Link](#)

- Built an end-to-end retail analytics workflow for 8,399 order lines and 5,496 orders, covering data cleaning, star-schema modelling, SQL analysis and BI reporting.
- Answered eight business questions with SQL joins, CTEs, subqueries and window functions; found that 4,264 order lines (50.8%) were loss-making despite a 10.2% overall profit margin.
- Identified furniture profitability issues, discount-driven margin erosion and shipping delays, and translated the findings into actionable business recommendations.
- Presented sales, customer, product and shipping insights in Power BI, Tableau and Excel dashboards.

UCI Online Retail II Customer Analytics and Predictive Modelling [Link](#)

- Cleaned and analysed 525,461 raw transaction rows down to 518,596 validated rows using data quality checks, classification and feature engineering.

- Compared repeat and one-time customer order value with Welch's t-test and the Mann-Whitney U test, supported by confidence intervals and effect size analysis.
- Benchmarked Logistic Regression, Decision Tree and Random Forest models for cancellation prediction with 5-fold cross-validation; selected Random Forest (depth 20) with 0.8448 ROC-AUC.
- Adjusted the decision threshold to 0.60, achieving 31.63% precision, 30.16% recall and an F1 score of 0.3088 for cancellation detection.

Solar Active-Region Machine Learning Benchmark [Link](#)

- Built a reproducible machine learning benchmark on a published solar active-region catalogue for flare-productivity classification and regression.
- Evaluated models with cross-validation, ROC-AUC, PR-AUC, Brier score, R^2 , MAE and RMSE, and interpreted predictions with SHAP feature attribution.
- Found maximum magnetic flux as the strongest aggregate predictor while keeping interpretation separate from out-of-sample performance claims.
- Designed a follow-up time-resolved modelling approach with strict data-leakage controls for operational forecasting.

EDUCATION

M.Sc. Physics in Physics, National Institute of Technology Manipur (GPA: CGPA: 8.62/10.0) '20 — '22
India

- Relevant coursework: Quantum Mechanics, Statistical Physics, Numerical Methods, Computational Physics.

B.Sc. Physical Science in Physical Science, University of Delhi (GPA: CGPA: 7.82/10.0) '15 — '18
India

- Relevant coursework: Quantum Mechanics, Statistical Physics.

AWARDS

Selected as Junior Research Fellow for DST-funded research

National Institute of Technology Delhi

Selected as a Junior Research Fellow for DST-funded research at National Institute of Technology Delhi.

Selected as Junior Research Fellow for MoE-funded research

Indian Institute of Astrophysics

Selected as a Junior Research Fellow for MoE-funded research at the Indian Institute of Astrophysics.

First author on a research manuscript

First author of a manuscript prepared for peer-reviewed publication.

Presented research findings

IIA In-house Symposium

Presented research findings at the IIA In-house Symposium in Bengaluru.

CERTIFICATIONS

Graduate Aptitude Test in Engineering (GATE) – Physics Jan '22

ADDITIONAL ANALYTICS EXPERIENCE

- Applied Excel for Pivot Tables, VLOOKUP/XLOOKUP, data cleaning, and KPI reporting.
- Analysed website and digital performance metrics using Google Analytics.
- Built visualisations in Matplotlib, Seaborn, and Plotly, and developed analytics apps in Streamlit.
- Queried and analysed relational data in MySQL and PostgreSQL.